

AdaptiveSats: Improving Institutional Bitcoin Accumulation Strategies

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1 Executive Summary

In partnership with the Trilemma Foundation, this project evaluates whether public Bitcoin (Nakamoto (2008)) market and on-chain data can improve long-term Bitcoin accumulation compared with naive Dollar-Cost Averaging (DCA), which serves as our primary baseline strategy. This supports Trilemma’s goal of building an open-source framework for evaluating Bitcoin accumulation strategies. Unlike naive DCA, which invests a fixed amount at regular intervals, our adaptive DCA strategy adjusts allocations based on public signals and changing Bitcoin market conditions. The goal is to improve long-term accumulation efficiency by acquiring more Bitcoin with the same fixed budget. We measure this using Sats per Dollar (SPD), which represents the amount of Bitcoin or satohis (Sats) accumulated per dollar invested. Using the BRK (Bitcoin Research Kit Contributors (2025)) metrics dataset from the StackSats ecosystem, which contains daily market and on-chain metrics from 2009 to 2026, we analyze 41,000+ metrics, apply feature reduction, and backtest adaptive allocation strategies against naive DCA.

2 Introduction

Long-term Bitcoin accumulation is important for institutions and foundations building Bitcoin exposure. One common approach is Dollar-Cost Averaging (DCA), where a fixed amount is invested regularly. We use naive DCA as the baseline, but it has limitations because it does not adjust to changing market conditions. For institutions with fixed budgets, this may reduce efficiency by missing favorable buying periods.

This problem is important because Bitcoin’s public and auditable data allows smaller institutions to explore data-driven accumulation strategies without relying on privileged market access.

We evaluate whether public Bitcoin market and on-chain data can support an adaptive DCA strategy that improves SPD relative to naive DCA while using the same fixed accumulation budget. After initial analysis, we develop and backtest adaptive allocation strategies against naive DCA. The final product will include a cleaned data pipeline strategy, backtesting logic, and evaluation results comparing adaptive accumulation to naive DCA for the Trilemma Foundation and the open-source community.

2.0.1 Objectives

- Identify on-chain/market signals predicting favorable accumulation windows

- Build and validate an adaptive daily allocation model

3 Data & EDA

Table 1: Dataset overview

Property	Value
File size	~1.08 GiB
Total rows	236,259,020
Daily observations	6,274 days
Distinct metric keys	41,407
Date range	2009-01-03 to 2026-03-13

The 41,407 metrics span several categories including price, market valuation, holder behaviour, network activity, and technical indicators.

3.0.1 Identifying Feature Families

Family name ▼	Count of features
UTXO Age Cohorts	18,342
Price	6,050
Address Distribution by Balance	5,250
Age & Halving Cohorts	2,737
Entities	2,368
Address Activity by Tech Type	1,732
Pool-Specific Economics	1,612
Network Activity	1,415
Holder Cohorts	842
Market & Valuation	370
Benchmarks & DCA	369
Supply & Scarcity	130
Profitability & SOPR	117
Block Reward Distributions	27
Technical Indicators	25
Metadata	22
Grand Total	41,408

Figure 1: Feature count by family

3.0.2 Zero Variance Features Selection

One of the first strategies we used in the feature selection step was removing columns with zero variance. According to Kuhn and Johnson (2019), features with a single unique value (zero variance) provide no information gain. Therefore, removing them helps avoid numerical instability.

After applying this approach, 2,810 features were removed from the training dataset.

3.0.3 Halving Events and Price History

Table 2 lists each halving date and the resulting block reward reduction. Because Bitcoin has a hard cap of 21 million coins, the halving reinforces its programmed scarcity. Figure 2 shows how Bitcoin’s price (market cap / circulating supply) has evolved across those epochs, illustrating events are generally followed by significant upward price trends. (Subawa et al. (2025))

Table 2: Bitcoin halving events and block rewards

Event	Date	Block reward
1st halving	28 November 2012	25 BTC
2nd halving	9 July 2016	12.5 BTC
3rd halving	11 May 2020	6.25 BTC
4th halving	20 April 2024	3.125 BTC

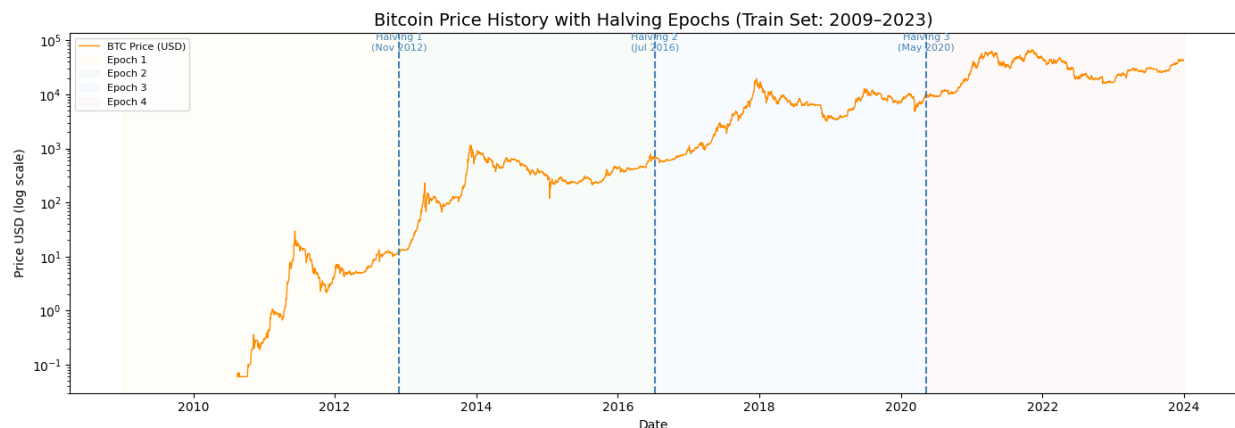


Figure 2: Bitcoin price (market cap / circulating supply) with halving epochs annotated

MVRV, NUPL, and SOPR are key on-chain metrics that have historically signaled favorable accumulation opportunities when they indicate undervaluation (e.g., $MVRV < 1$, $NUPL < 0$, $SOPR < 1$); see Section 7 for full metric definitions. Figure 3 shows how these metrics have varied across different market cycles, with lower values often coinciding with cycle bottoms and prime accumulation zones.

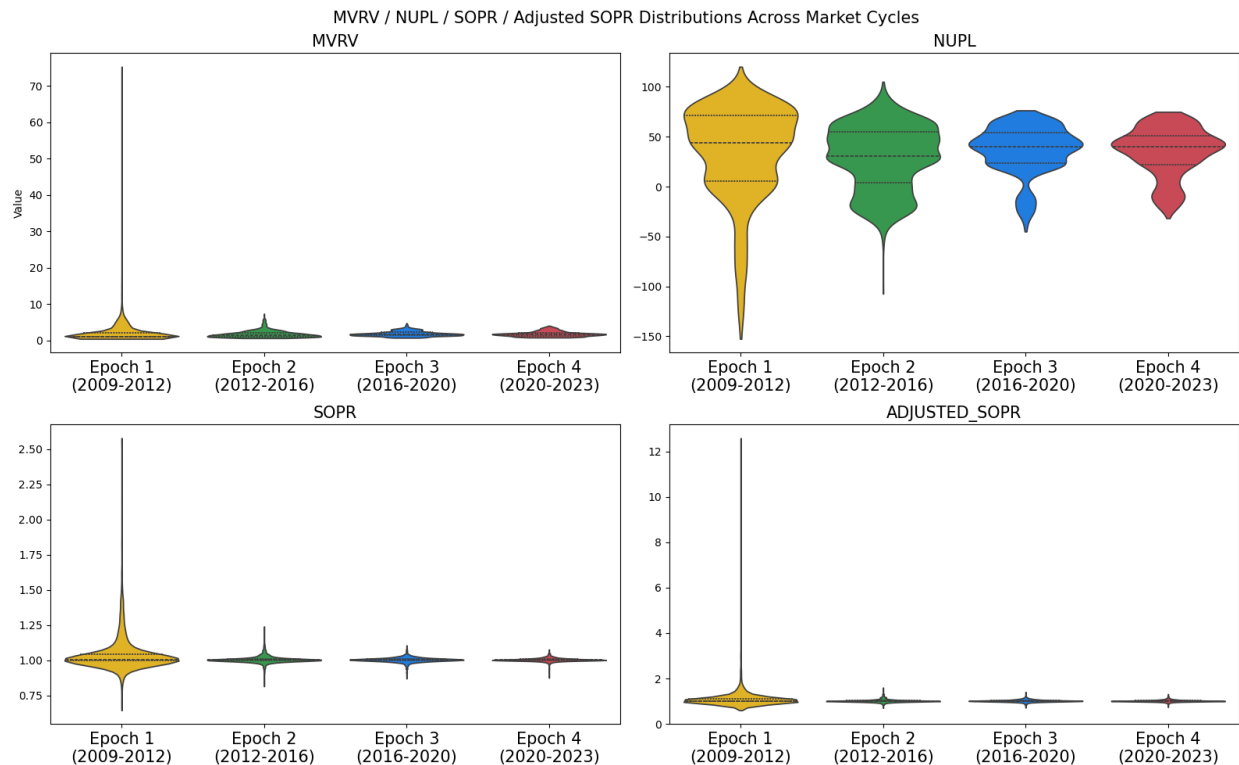


Figure 3: MVRV / NUPL / SOPR distributions across market cycles

- **Limitations:** The dataset requires lazy loading due to its size. Many metrics only begin after 2009, and there is heavy multicollinearity across the 41k features.

3.0.4 Clustering and VIF

After identifying the feature families in our data, we performed hierarchical clustering on each family because we believe there would be large similarities among the members. For each cluster identified, we plan to calculate the VIF, to give us an understanding of the level of multicollinearity present in the data.

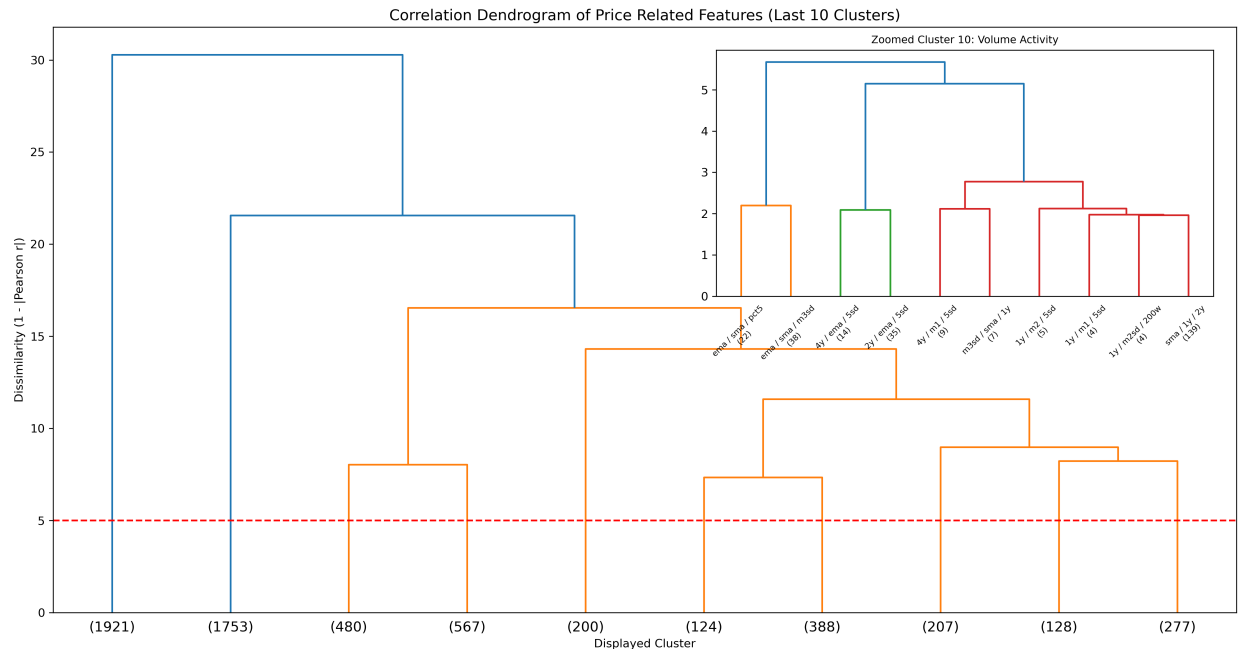


Figure 4: Correlation Dendrogram of Price Features

4 Data Science Approach

4.0.1 Feature Reduction

With over 41,000 features, we aim to identify which features capture similar information to make the dataset more manageable for modeling. To achieve this, we follow a structured feature reduction approach.

First, we group features based on their names using rule based parsing, which helps create feature families such as Price, Benchmark & DCA related features, as depicted in Figure 1. Next, within each family, we apply hierarchical clustering to group features that behave similarly over time, using dendrograms for visualization. Then, from each cluster, we retain a small set of important features and remove those that are highly similar, reducing the dataset size while preserving key information.

This process is computationally intensive when run locally with limited resources. We may explore efficient feature reduction methods in future iterations.

4.0.2 Baseline Strategies

To evaluate our approach, we aim to implement a **Naive Dollar-Cost Averaging (DCA)** strategy as the baseline, where a fixed amount is invested at regular intervals. Our goal is to outperform it by improving accumulation efficiency.

To improve upon this baseline, we are exploring more effective accumulation strategies. In particular, we plan to implement **Value Averaging** as our primary strategy, which adjusts investment based on a target portfolio growth path, allocating more capital during price declines and less during price increases. This aligns well with our objective of maximizing **Sats per Dollar (SPD)**.

In addition, we may also evaluate a **200-day SMA (Simple Moving Average) based strategy**, which adjusts investment based on long-term market trends, providing a simple signal-based benchmark.

4.0.3 Model Development

Our proposed strategy will leverage the reduced feature set to design a data-driven accumulation approach.

The objective is to:

- Identify periods of undervaluation
- Dynamically adjust investment allocation
- Improve accumulation efficiency compared to the defined baseline strategies

4.0.4 Train/Test Split

To avoid look-ahead bias and reflect a realistic backtesting setup, we use a chronological train/test split.

- **Training set:** Data up to December 2023
- **Test set:** January 2024 to March 2026

The test set will remain untouched during model development and will only be used for final evaluation.

4.0.5 Future Work

In future iterations, we may explore additional model such as LSTM (Long Short-Term Memory) based model. This approach will be evaluated against the baseline strategies to assess potential improvements in accumulation efficiency.

5 Evaluation & Success Criteria

5.0.1 Backtesting

The key metric used in backtesting was Sats-per-dollar (SPD). The backtesting process included two frameworks: Validation Checks and Performance Results HyperTrial (2025):

Validation Checks included integrity checks designed to reduce leakage risk and arithmetic drift. These were evaluated through:

1. **Weight Sum Validation:** Confirms that each window’s daily allocations sum to exactly 100% of the budget within floating-point tolerance, ensuring a fair comparison against the naive DCA baseline.
2. **Forward-Leakage Test:** Future data is masked and the model weights are recomputed to ensure the results remain consistent.
3. **Win-rate Requirement:** The model must outperform naive DCA in more than 50% of the evaluation windows.

Performance Results summarized the key SPD and percentile results:

1. **Percentile Statistics:** Measures the minimum, maximum, mean, median, and exponentially weighted percentile (where recent windows receive greater weight) of the winning rate against naive DCA.
2. **Strategy Validation:** Evaluates whether the model outperforms naive DCA based on the win-rate threshold (greater than 50%), using 2558 rolling one-year windows from 2018 to the present.

The model is then scored by combining the win rate and exponentially weighted percentile into a single benchmark.

5.0.2 Metric

- **Primary Metric (Sats per Dollar - SPD):**

We evaluate accumulation efficiency using Sats per Dollar (SPD), defined as the total Bitcoin accumulated per unit of USD invested. A higher SPD indicates better performance, as it reflects a lower average acquisition cost.

5.0.3 Success Criteria

A strategy is successful if it achieves higher SPD than the naive DCA baseline in more than 50% of tested rolling windows, is statistically significant ($p < 0.05$), holds across at least one bull and one bear cycle, passes all strict validation checks mentioned above, and has no look-ahead bias.

5.0.4 Assumptions and limitations

This project assumes public Bitcoin data contains useful accumulation signals, but backtesting may not reflect future market behavior, feature reduction may remove useful signals, and real-world constraints may be simplified. Results should therefore be viewed as a controlled evaluation, not guaranteed performance

5.0.5 Stakeholders

- **Trilemma Foundation:** Interested in improving long-term Bitcoin accumulation efficiency
- **Open-source community:** Focused on transparency, reproducibility, and practical usability of the strategy

6 Timeline

Table 3: Project timeline with key milestones

Milestone	Target Date
Train/test split + EDA scripts	May 15, 2026
Feature reduction + baseline DCA	May 22, 2026
ML model + walk-forward CV	May 29, 2026
Backtesting + final evaluation	June 12, 2026
Final report + model	June 23, 2026

7 Appendix 1: On-Chain Metrics for Long Accumulation Identification

The following table summarises the key on-chain and market metrics and their role in identifying favourable Bitcoin accumulation windows.

Table 4: Key on-chain metrics and their role in long accumulation identification

Metric	Expands to	Definition	Accumulation Signal
mvr	Market Value to Realized Value	Ratio of market capitalisation to realised capitalisation (the aggregate cost basis of all coins).	Values < 1 (market price below aggregate cost basis) historically mark cycle bottoms and prime accumulation zones.
nupl	Net Unrealized Profit/Loss	$\text{NUPL} = \frac{\text{Unrealized Profit} - \text{Unrealized Loss}}{\text{Market Cap}};$ measures the net paper gain/loss of all holders as a fraction of market cap.	Values < 0 indicate the majority of holders are in loss.
sopr	Spent Output Profit Ratio	Ratio of the USD value of coins at the time they are spent (sold) to their USD value when they were created (cost basis). $\text{SOPR} = \frac{P_{\text{sell}}}{P_{\text{buy}}}$	SOPR < 1 means coins are being sold at a loss on aggregate, indicating capitulation.
adjusted_sopr	Adjusted SOPR	SOPR calculated after excluding very short-term (< 1 hour) spends to remove noise from exchange transfers and re-organisations.	Same interpretation as SOPR but more reliable.
realized_price	Realized Price	Average USD price at which every coin was last moved on-chain; represents the aggregate cost basis per coin. $\text{RP} = \frac{\text{Realized Cap}}{\text{Supply}}$	When market price is below Realized Price, most holders are at an unrealized loss, which has historically signaled favorable accumulation opportunities.

8 Appendix 2 : Glossary of Terms

- **Bitcoin (BTC):** A decentralized digital currency that operates on a peer-to-peer network without a central authority. It was created in 2009 by an anonymous person or group using the pseudonym Satoshi Nakamoto (Nakamoto (2008)).
- **On-chain data:** Information recorded on the blockchain, such as transactions, addresses, and balances, which is publicly accessible and can be analyzed to derive insights about market behavior and participant activity.
- **Dollar-Cost Averaging (DCA):** An investment strategy where a fixed amount of money is invested at regular intervals, regardless of the asset's price, to reduce the impact of volatility over time.

- **Naive DCA:** A simple implementation of DCA where the same amount is invested at each interval without any adjustments based on market conditions or signals.
- **Sats per Dollar (SPD):** A metric that measures the amount of Bitcoin (in satoshis) accumulated per unit of USD invested, used to evaluate the efficiency of an accumulation strategy.
- **Halving:** A programmed event in the Bitcoin protocol that occurs approximately every four years, reducing the block reward for miners by half, which historically has had significant impacts on Bitcoin’s price and market dynamics.
- **Backtesting:** The process of testing a trading strategy or model on historical data to evaluate its performance and robustness before deploying it in live markets. Includes validation checks to ensure no violations and look-ahead bias and that the strategy is evaluated fairly against a baseline.
- **Hidden Markov Models (HMMs):** Hidden Markov models are probabilistic models in which observations, in this case prices or market regimes, are dependent on a latent (hidden) Markov process. They are used to represent systems that transition between hidden states over time, while producing observable outputs.
- **Markov process:** A Markov process is a stochastic (random) mathematical process in which future outcomes depend solely on the present state, not on the path of past events. It operates on the rule that “the future is independent of the past, given the present.”
- **Simple Moving Average (SMA):** A simple moving average (SMA) is a tool used in financial analysis to determine the average price of an asset over a set number of periods, typically calculated using closing prices. It smooths out price fluctuations and aids investors in identifying potential trends.
- **Variance Inflation Factor (VIF):** A measure of multicollinearity in regression analysis, indicating how much the variance of an estimated regression coefficient increases due to collinearity among the predictors. A VIF value greater than 5 or 10 is often considered indicative of high multicollinearity.
- **Long Short Term Memory (LSTM):** A type of recurrent neural network (RNN) architecture that is capable of learning long-term dependencies in sequential data, often used in time series forecasting and natural language processing tasks.
- **Cross Validation (CV):** A technique for assessing the generalizability of a predictive model by partitioning the data into subsets, training the model on some subsets and validating it on others, to ensure that the model performs well on unseen data.

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